

BSc(Surveying) | RECO40II Dissertation

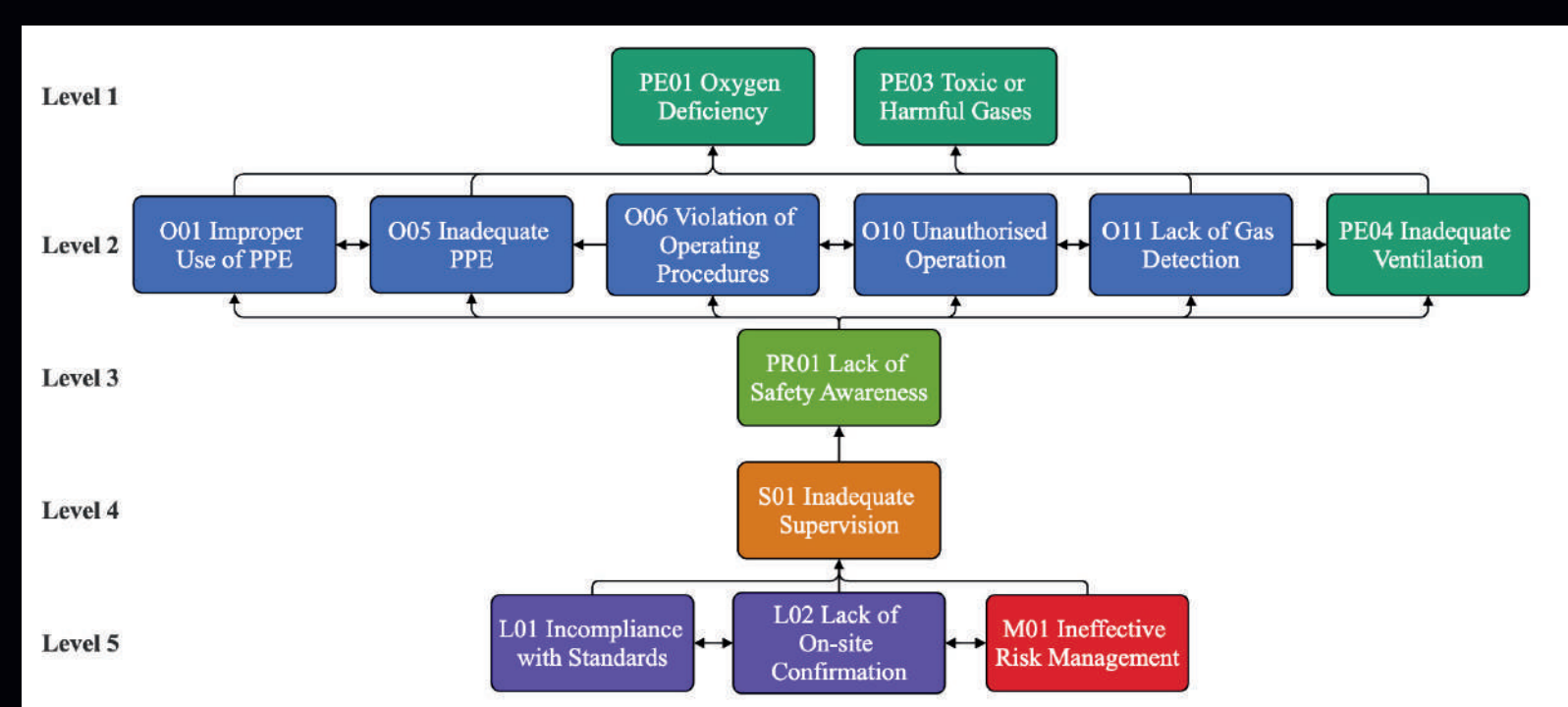
Mapping the Critical Factors of Confined Space Accidents in the Built Environment in Hong Kong: An Interpretive Structural Modelling (ISM) Study

STUDENT: Cheung Lok Hang

SUPERVISOR: Prof. Frank Xue

Major Findings

1. Fourteen critical factors for confined space accidents in Hong Kong's AEC industry were confirmed, with Physical and Operational factors rated as the most critical.
2. Interpretive Structural Modelling analysis revealed a five-level hierarchy, with Legal & Policy factors and Management factors as the most fundamental drivers of confined space accidents.
3. Three targeted safety interventions were proposed regarding gaps identified in tackling key drivers.



Abstract

Confined space operations in the built environment represent one of the most dangerous occupational environments. In Hong Kong, accidents in these settings have accounted for a significant proportion of occupational fatalities over the past decade. A review of international literature reveals gaps in understanding the interrelationships among critical factors contributing to confined space accidents. To address this gap, a two-step mixed-methods research design was adopted. First, a Likert-scale questionnaire survey was developed following a comprehensive literature review of critical accident causation factors to identify and rank the key risks. Subsequently, semi-structured interviews were conducted to gather expert insights, which were then used to map the driving and dependent interrelationship network between the identified factors using Interpretive Structural Modelling (ISM) and Matrix d'Impacts Croisés Multipliée Appliquée à un Classement (MICMAC) analysis.



The questionnaire survey confirmed 14 critical factors, with Physical and Operational factors identified as the most critical. The ISM analysis revealed a five-level hierarchical network of these 14 factors, where Legal and Policy factors and Management factors were found to be the most fundamental drivers of confined space accidents. MICMAC analysis identified Supervision and Personal Readiness factors as critical linkages to confined space accidents, leading to direct outcomes such as physical hazards. The findings further reveal that, despite the 2024 revision of the Code of Practice for Safety and Health at Work in Confined Spaces (CoP), significant gaps remain in risk management, supervision, and behavioural practices. Based on the findings, three safety interventions were proposed.

The study provides both theoretical findings and practical suggestions on the driver-dependency relationships of critical factors for confined space accidents in the Hong Kong context. The recommended interventions aim to break critical causal pathways, reduce accident severity and promote a safer occupational confined space environment for the AEC industry in Hong Kong.